

Dr. Xiao (Brandon) Han

(+44) 07529989025
brandon.hxiao@gmail.com
brandonhan.uk

Senior AI Research Scientist, Meta
London, United Kingdom

RESEARCH INTERESTS

My research lies at the intersection of **multimodal representation learning** and **generative AI**. I am interested in learning transferable vision-language representations that connect images, video, text, and other visual modalities, with applications spanning cross-modal and compositional retrieval, multimodal adaptation, and fine-grained visual understanding. Building on this foundation, my current work focuses on **controllable image and video generation**, including reference-conditioned generation and creative 2D/3D content synthesis. My long-term goal is to develop unified multimodal systems that combine strong understanding with high-quality, controllable generation for real-world creative applications.

EXPERIENCE

Meta (Facebook UK Limited)

Senior AI Research Scientist

AI Research Scientist

London, UK

Jul. 2025 – Present

Oct. 2023 – Jul. 2025

- Team: **Monetization Generative AI (MGenAI) London**, within Core Ads Growth (CAG).
- Core contributor to **monetization-focused media generation** across Meta Ads platforms, including Ads Manager, Facebook, and Instagram.
- Lead and contribute to the **end-to-end development of in-house video-generation models** tailored to advertising, including **image animation**, **reference-to-video generation**, and **talking-avatar generation**.
- Lead and contribute to **creative-image generation for ads**, including image variants and background generation, using diffusion models deployed at **million-scale daily volume**.
- Work across the **full model stack**: data curation, pre-training, supervised fine-tuning (SFT), reinforcement learning (RL), distillation, inference optimization, evaluation, and **agentic-system integration** for prompt generation and rewriting, user-intent understanding, and automated video-quality assessment.

University of Surrey

Postgraduate Researcher

Guildford, UK

Jan. 2021 – Oct. 2023

- Researched **multimodal deep learning**, with a focus on generative modeling and representation learning.
- Awarded a **University of Surrey FEPS/iFlyTek Ph.D. Studentship**, including a tuition-fee waiver and 36 months of maintenance funding.
- Published **six top-tier papers** and presented the work at international academic conferences.
- Served as a demonstrator for undergraduate and postgraduate courses.

PixelShift.AI

Computer Vision Algorithm Intern

Shanghai, China

May 2020 – Sep. 2020

- Mentors: Zhiming Ma and Meng Zhang.
- Developed an **immersive AR application** using Google MediaPipe and deployed generative models.

EDUCATION

University of Surrey

Doctor of Philosophy (Ph.D.)

Guildford, UK

Jan. 2021 – Jul. 2024

- Major in “Vision, Speech and Signal Processing” at the Centre for Vision, Speech and Signal Processing.
- Supervisors: Prof. Tao Xiang and Prof. Yi-Zhe Song.
- Thesis: **Fine-Grained Vision-Language Learning**.
- Examiners: Prof. Andrea Vedaldi (University of Oxford & Meta) and Dr. Anjan Dutta (internal).

- Major in “Information Engineering” at the College of Information Science and Electronic Engineering.
- Cumulative GPA: 3.92 / 4.00 (88.13 / 100); Third-year GPA: 3.94 / 4.00 (89.46 / 100).
- Thesis: Deep Learning-Based Features Prediction for Mass Spectrometry of Protein.

Non-Degree Academic Experiences

- Visiting student with Prof. Li Zhang @ ZVG, Fudan University 2020 Winter · Shanghai, China
- Research assistant with Dr. Changbin Yu @ AiR, Westlake University 2019–2020 · Hangzhou, China
- Visiting student with Prof. L. Jay Guo @ EECS, University of Michigan 2019 Summer · Ann Arbor, MI, USA
- Exchange student with iESR program @ University of Notre Dame 2018 Summer · South Bend, IN, USA

SELECTED PUBLICATIONS

(**Bold** indicates Xiao (Brandon) Han; * equal contribution, [†] corresponding author. Full list on [Google Scholar](#).)

Conference

1. *VecGlypher: Unified Vector Glyph Generation with Language Models*
Xiaoke Huang, Bhavul Gauri, Kam Woh Ng, Tony Ng, Mengmeng Xu, Zhiheng Liu, Weiming Ren, Zhaochong An, Zijian Zhou, Haonan Qiu, Yuyin Zhou, Sen He, Ziheng Wang, Tao Xiang, **Xiao Han**[†].
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026.
2. *Scaling Zero-Shot Reference-to-Video Generation*
Zijian Zhou, Shikun Liu, Haozhe Liu, Haonan Qiu, Zhaochong An, Weiming Ren, Zhiheng Liu, Xiaoke Huang, Kam Woh Ng, Tian Xie, **Xiao Han**, Yuren Cong, Hang Li, Aditya Patel, Tao Xiang, Sen He.
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026.
3. *TUNA: Taming Unified Visual Representations for Native Unified Multimodal Models*
Zhiheng Liu, Weiming Ren, Haozhe Liu, Zijian Zhou, Shoufa Chen, Haonan Qiu, Xiaoke Huang, Zhaochong An, Yukang Yang, Aditya Patel, Tony Ng, **Xiao Han**, Ding Liu, Juan-Manuel Perez-Rua, Sen He, Jürgen Schmidhuber, Wenhui Chen, Ping Luo, Wei Liu, Tao Xiang, Jonas Schult, Yuren Cong.
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026. [Highlight, Top 3.5%]
4. *Learning Flow Fields in Attention for Controllable Person Image Generation*
Zijian Zhou, Shikun Liu, **Xiao Han**, Haozhe Liu, Kam Woh Ng, Tian Xie, Yuren Cong, Hang Li, Mengmeng Xu, Juan-Manuel Perez-Rua, Aditya Patel, Tao Xiang, Miaoqing Shi, Sen He.
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025.
5. *HeadSculpt: Crafting 3D Head Avatars with Text*
Xiao Han^{*}, Yukang Cao^{*}, Kai Han, Xiatian Zhu, Jiankang Deng, Yi-Zhe Song, Tao Xiang[†], Kwan-Yee K. Wong[†].
Neural Information Processing Systems (NeurIPS), 2023.
6. *Controllable Person Image Synthesis with Pose-Constrained Latent Diffusion*
Xiao Han, Jiankang Deng, Xiatian Zhu, Yi-Zhe Song, Tao Xiang.
International Conference on Computer Vision (ICCV), 2023.
7. *FAME-ViL: Multi-Tasking Vision-Language Model for Heterogeneous Fashion Tasks*
Xiao Han, Xiatian Zhu, Licheng Yu, Li Zhang, Yi-Zhe Song, Tao Xiang.
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023. [Highlight, Top 2.5%]
8. *FashionViL: Fashion-Focused Vision-and-Language Representation Learning*
Xiao Han, Licheng Yu, Xiatian Zhu, Li Zhang, Yi-Zhe Song, Tao Xiang.
European Conference on Computer Vision (ECCV), 2022.
9. *UIGR: Unified Interactive Garment Retrieval*
Xiao Han, Sen He, Li Zhang, Yi-Zhe Song, Tao Xiang.
CVPR Workshop (CVPRW), 2022.
10. *Text-Based Person Search with Limited Data*
Xiao Han, Sen He, Li Zhang, Tao Xiang.
British Machine Vision Conference (BMVC), 2021.

Journal

11. *MarDini: Masked Auto-regressive Diffusion for Video Generation at Scale*
Haozhe Liu, Zijian Zhou*, Shikun Liu*, Mengmeng Xu, Yanping Xie, **Xiao Han**, Juan C. Pérez, Ding Liu, Kumara Kahatapitiya, Mengling Jia, Jui-Chieh Wu, Sen He, Tao Xiang, Juergen Schmidhuber, Juan-Manuel Pérez-Rúa.
Transactions on Machine Learning Research (TMLR), 2025.

HONORS & AWARDS

- 2022 · **2nd Place**, eBay eProduct Visual Search Challenge – FGVC9 (CVPR 2022)
- 2021–2024 · Faculty of Engineering and Physical Sciences / iFlyTek Scholarship, Univeristy of Surrey
- 2019 · Chunzhen International Exchange Scholarship, Zhejiang University
- 2017–2019 · **3rd Prize**, Academic & Outstanding Student Scholarship, Zhejiang University
- 2018 · Yongping Scholarship, Zhejiang University
- 2018 · **3rd Prize**, National Talent Training Base Scholarship, Zhejiang University
- 2018 · **Honorable Mention**, Mathematical Contest in Modeling (MCM)
- 2017 · **3rd Prize**, Physics Innovation Competition in Zhejiang Province (Theory)

MENTORSHIP & ADVISING

Meta AI – Research Scientist Intern Manager

Hosting and advising visiting Ph.D. research scientist interns

London, UK
Aug. 2024 – Present

- **Kam Woh Ng** (Ph.D. at University of Surrey; Summer 2024).
Generative neural rendering and video generation (Kaleido, ICLR 2026).
- **Xiaoke Huang** (Ph.D. at UC Santa Cruz; Summer 2025).
Training language models for vector glyph generation (VecGlypher, CVPR 2026).

TEACHING

University of Surrey

Teaching Assistant / Demonstrator and Project Tutor

Guildford, UK
2023 Spring

- **EEEM071 – Advanced Topics in Computer Vision and Deep Learning** (under **Prof. Yi-Zhe Song**).
Responsible for the final coursework design (Vehicle ReID), lab tutorial preparation (fundamentals for NumPy / PyTorch), demonstrating in lab sessions with 145+ PG/T students having mixed backgrounds, and marking the coursework reports.
- **EEEM004 – 60 Credit Standard Project** (under **Dr. Xiatian Zhu**).
Responsible for guiding PG/T final projects “Zero-/Few-shot learning with language supervision”.

ACADEMIC SERVICE

Conference Reviewer (31 papers reviewed)

- International Conference on Learning Representations (ICLR) 2024, 2025 (×2).
- Conference on Neural Information Processing Systems (NeurIPS) 2023 (×3).
- IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2022, 2023, 2024, 2026 (×7).
- International Conference on Computer Vision (ICCV) 2023 (×3).
- European Conference on Computer Vision (ECCV) 2022, 2024, 2026 (×6).
- AAAI Conference on Artificial Intelligence (AAAI) 2023, 2024 (×8).
- CVPR Workshop on Computer Vision for Fashion, Art, and Design (CVFAD) 2023 (×1).
- IEEE International Conference on Multimedia and Expo (ICME) 2022 (×1).

Journal Reviewer (6 papers reviewed)

- IEEE Transactions on Visualization and Computer Graphics (TVCG) (×1).
- Computer Graphics Forum (CGF) (×1).
- IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI) (×2).
- IEEE Transactions on Neural Networks and Learning Systems (TNNLS) (×1).
- IEEE Transactions on Circuits and Systems for Video Technology (TCSVT) (×1).